26 Hillforts and Pastoral Landscapes in the Fucino Basin: A GIS Approach to Transhumance in the Central Apennines

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This paper is framed within the European project E-TransAp: Transhumance Pastoralism in the Apennines, which investigates the long-term relationship between mobile pastoral practices and the occupation of upland landscapes in central Italy. Focusing on the Fucino basin and its surrounding mountain ranges, the study examines the role of hillforts in the organisation and control of pastoral resources. As a first step, all known hillforts in the Fucino area were systematically mapped within a QGIS environment, drawing on published surveys, archival data and remote sensing. This dataset provides a comprehensive overview of fortified upland sites in the region and allows their topographic setting, visibility and spatial distribution to be analysed consistently. Subsequently, a land-use analysis was carried out to identify areas suitable for pastoral exploitation. Environmental variables such as elevation, slope, aspect, vegetation cover and water availability were combined to model potential grazing areas, distinguishing between seasonal upland pastures and lower-lying winter grounds. The results reveal a significant spatial relationship between hillforts and pastoral landscapes. Many fortified sites are located on the margins of high-quality grazing areas or at strategic points overlooking access to them. This suggests that hillforts may have played a key role in controlling and protecting pastoral resources, rather than functioning solely as defensive or residential centres. Finally, the spatial data were integrated with known transhumance routes, including the Celano - Foggia tratturo and secondary droving paths. This comparison highlights a close correspondence between the location of hillforts and the main axes of livestock mobility, pointing to a long-lasting structuring of the landscape around pastoral movement. Overall, the paper argues that pastoralism was a central factor in the emergence and distribution of hillforts in the central Apennines.